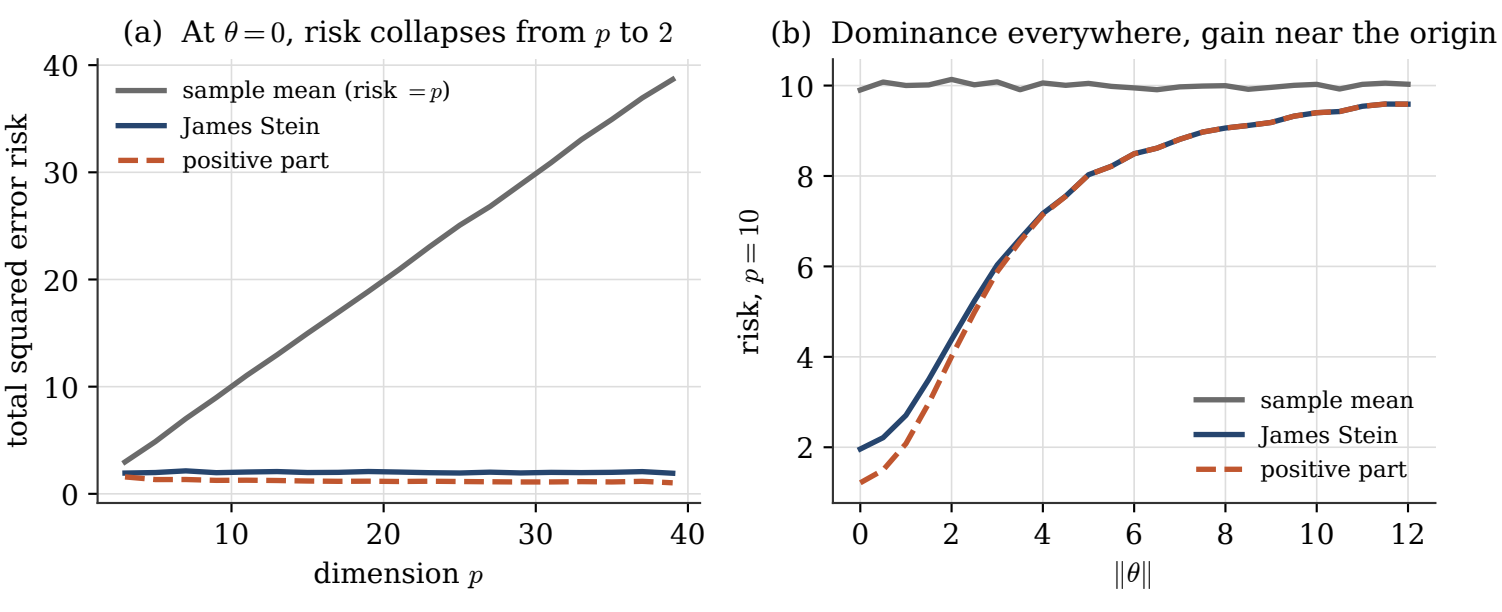
(a) At $\theta = 0$, risk collapses from p to 2
sample mean (risk $= p$)
James Stein
positive part
total squared error risk
dimension p
0
10
20
30
40
(b) Dominance everywhere, gain near the origin
sample mean
James Stein
positive part
risk, $p = 10$
$\|\theta\|$
2
4
6
8
10
12